If the live demo fails

Keep this file and the contingency PDF open before the talk. This page is self-contained: no scripts, server, network, or installed fonts required.

1. For a failed edit: Reset, then Run once. If it still fails, switch to the extracted replay at `http://127.0.0.1:7342`.
2. For internet/ngrok failure: use the local replay. Start it with `python3 -m http.server 7342 --bind 127.0.0.1` inside the extracted ZIP.
3. For server, Java, or browser-demo failure: open this HTML file directly, or show the corresponding section of `lambda-world-contingency.pdf`. All numbers below are recorded Clojure results, rounded for display.
4. If the projector itself fails, use these same statements as the spoken explanation; do not spend the talk debugging the room.

01 / Formation macro + gradient descent

Trigger: Macro expansion or live evaluation fails.

Replay route: chapter #flock, step 0; press S for stage and T for the calculation.

Show this if the live demo is unavailable

```
(f/fleet [⛵ · · ⛵]
         [· · ⛵ ·])
;; => [0.0 0.0 3.0 0.0 2.0 1.0]

(take 101 (iterate step {:x boats}))
```

Recorded boat 1: x	`[0, 0]`
−rate × gradient	`[0.15, -0]`
Next x	`[0.15, 0]`

Say

The macro produces ordinary coordinates. Descent returns State → State; iterate supplies the history. The first movement is visible as vector addition even without animation.

> **Keep the claim precise:** These are candidate formations, not physical boat dynamics.

02 / A penalty decorates the objective

Trigger: The boats do not animate or the edited penalty cell fails.

Replay route: chapter #constraints, step 100; press S for stage and T for the calculation.

Show this if the live demo is unavailable

```
augmented(x) = f(x) + weight × Σ max(0, gᵢ(x))²

(h/penalize objective constraints 3.0)
```

Final distance cost	6.8067
Weighted penalty	1.6112
Closest pair	0.8332
Desired separation	1.1

Say

The optimizer still receives one function. Positive violations add cost; the loop stays the same. Point to the closest distance to show the compromise.

> **Keep the claim precise:** A finite soft penalty does not guarantee feasibility or collision avoidance.

03 / Momentum decorates the step

Trigger: The beta edit, comparison control, or evaluator fails.

Replay route: chapter #momentum, step 1; press S for stage and T for the calculation.

Show this if the live demo is unavailable

```
v′ = βv + delta
x′ = x + v′

(with-momentum plain-step 0.75)
```

Boat 1 position	`[0.15, 0]`
Base movement	`[0.1425, -0]`
Carried movement	`[0.1125, 0]`
Next position	`[0.405, 0]`

Say

Add base movement and carried velocity to the current position. Beta zero removes the carry and recovers the base descent trajectory. Iterate receives the same function signature.

> **Keep the claim precise:** Momentum can overshoot; this is not an acceleration guarantee.

04 / Pass the execution strategy as a function

Trigger: pmap is slower, stalls, or the runtime is unavailable.

Replay route: chapter #parallel, step 0; press S for stage and T for the calculation.

Show this if the live demo is unavailable

```
(evaluate-population map  fitness population)
(evaluate-population pmap fitness population)

(vec (map-fn score population))
```

Recorded candidates	32
Recorded map / ms	173.5889
Recorded pmap / ms	141.6998
Identical ordered results	True

Say

Independent candidate trajectories can be mapped in parallel. Vec forces realization before timing. A slower pmap result illustrates overhead and is a valid outcome.

> **Keep the claim precise:** These are recorded timings from one run, not a benchmark or current measurement.

05 / Bellman transforms a value function

Trigger: The value landscape or live backup fails.

Replay route: chapter #bellman, step 0; press S for stage and T for the calculation.

Show this if the live demo is unavailable

```
Q(s,a) = Σ p × (reward + γV(next))
(T V)(s) = maxₐ Q(s,a)

(iterate improve initial-value)
```

Cell	`[5, 0]`
Old value	`0`
Best action	`[1, 0]`
Outcome 1: p × (r + γV)	`8.8`
Outcome 2: p × (r + γV)	`-0.009`
Outcome 3: p × (r + γV)	`-0.009`
New value	`8.782`

Say

Map the outcomes, reduce their weighted returns, then maximize across actions. Every calculation reads the same old value function.

> **Keep the claim precise:** This routes to a fixed confirmed location; heights are values, not probabilities.

06 / Belief update + two-search decision

Trigger: The probability map, Bayesian inspector, or planner display fails.

Replay route: chapter #pomdp, step 2; press Plan two searches.

Show this if the live demo is unavailable

```
(reductions (belief-step drift sensor)
            prior observations)

Q(b,a) = P(seen) + P(miss) × V₁(b′miss)
```

First search: west	0.8364
First search: central	0.7719
First search: east	0.7279

```
One optimal plan: west; if missed, east
0.4364 + 0.5636 × 0.7097 ≈ 0.8364
```

Say

After two recorded misses, compare the three hypothetical first actions. On another miss, condition the belief, predict another drift, then choose the second search. V_1 is the supplied continuation function.

> **Keep the claim precise:** Recorded observations are not an optimal-policy rollout. Travel cost is abstracted away; this planner is not proved by the Rainbow audit.

07 / Credit is a fold averaged over orders

Trigger: The arrival animation or attribution cell fails.

Replay route: chapter #shapley, step 23; press S for stage and T for the calculation.

Show this if the live demo is unavailable

```
1 − (1 − 0.8)² = 0.96

gain = value(team ∪ {boat}) − value(team)

(map #(arrival-trace value-of %)
     (permutations boats))
```

west	31.5152 pp
central	36.8582 pp
east	23.9127 pp
Total coverage	92.2861 pp

Say

A coalition-value function configures the fold. Each arrival receives its marginal coverage. Average all six orders to remove the ordering advantage.

> **Keep the claim precise:** This attributes expected coverage under one frozen belief, not causal credit for an observed rescue. Enumeration is factorial.

08 / Checked proof terms + a rejected term

Trigger: The kernel audit is slow, unavailable, or fails.

Replay route: chapter #rainbow, step 3; press S for stage and T for the calculation.

Show this if the live demo is unavailable

```
(kernel/check env proof-term claimed-type universes)

β = rate = gradient = 1
[x,v] = [0,2] → [1,1]

|V − W| ≤ ε  ⇒  |B(V) − B(W)| ≤ γε
```

Recorded declarations checked	11
Extra axioms	0
Number substituted for proof	rejected
Audit timestamp	2026-10-09T18:20:21.567487Z

Say

Show the recorded checked recurrence and Bellman bound, then the rejected number. State explicitly that these are previously checked exact terms, not a live kernel run.

> **Keep the claim precise:** No end-to-end proof of the floating-point program, renderer, POMDP, or general permutation Shapley implementation. Strict contraction requires $\gamma < 1$ and the theorem assumptions.

09 / Closing reveal

Trigger: The camera animation or rendering fails.

Replay route: chapter #overview, step 0; show the closing overview.

Show this if the live demo is unavailable

```
Objective → Objective
Step → Step
ValueFunction → ValueFunction

iterate · reductions · map
```

Applied Higher Order Functions · Jiri Knesl
Rainbow · proof terms as Clojure data

Say

The state spaces changed; the functional vocabulary stayed small. Find the state, the step, the functions configuring it, and the property worth checking.

> **Keep the claim precise:** Use this static overview and finish the talk. The camera is optional.

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